

Computer Networks

7.3 Analog Transmission



7.3 Analog Transmission

Why Analog Transmission?

Digital transmission needs a low-pass channel.

If only a **bandpass channel** is available, analog transmission is the choice.

Two Types of Conversion

- **Digital-to-Analog Conversion:**

Digital data → bandpass analog signal.

- **Analog-to-Analog Conversion:**

Low-pass analog signal → bandpass analog signal.

Key Idea

Modulation shifts information into a frequency range that matches the available channel.

Digital-to-Analog Conversion

Definition

Digital-to-analog conversion changes one characteristic of an analog signal according to the information contained in digital data.

A sine wave has three characteristics:

Amplitude

Frequency

Phase

Changing any one of these characteristics can represent digital data.

Digital-to-Analog Modulation Techniques

The three characteristics of a sine wave give three basic modulation techniques:

ASK

FSK

PSK

Fourth Technique

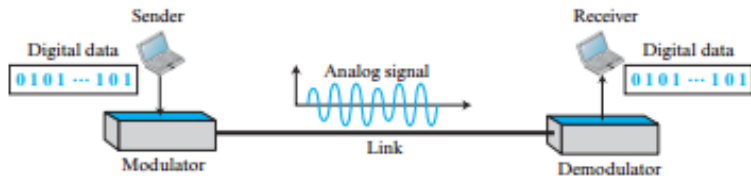
QAM changes both amplitude and phase.

Important

QAM is the most efficient of these options and is the mechanism commonly used today.

Figure 7.35 – Digital-to-Analog Conversion

Figure 7.35 *Digital-to-analog conversion*



Amplitude Shift Keying – ASK

Basic Idea

In ASK, the **amplitude** of the carrier is varied to represent digital data.

- Frequency remains constant.
- Phase remains constant.
- Amplitude changes according to the data.

Data → Amplitude Changes

ASK

Amplitude carries the information.

Binary ASK – BASK

ASK normally uses two amplitude levels.

This is called:

Binary ASK (BASK)

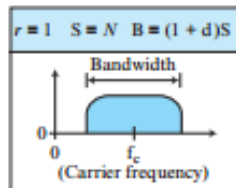
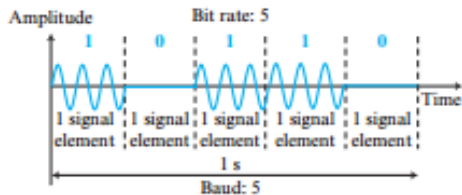
It is also called:

On-Off Keying (OOK)

- One signal level has peak amplitude 0.
- The other has the amplitude of the carrier.
- Frequency and phase remain constant.

Figure 7.36 – Binary ASK

Figure 7.36 *Binary amplitude shift keying*



ASK Bandwidth

ASK modulation produces a nonperiodic composite signal containing a continuous set of frequencies.

Bandwidth is proportional to the signal rate S .

$$B = (1 + d)S$$

where:

- B = bandwidth
- S = signal rate (baud)
- d = factor depending on modulation and filtering
- $0 \leq d \leq 1$

Therefore:

$$S \leq B \leq 2S$$

ASK and Bandpass Channels

The center of the ASK bandwidth is the carrier frequency:

$$f_c$$

Therefore, when a bandpass channel is available, we can choose f_c so that the modulated signal occupies the available bandwidth.

Major Advantage

Digital-to-analog conversion allows us to shift the resulting bandwidth to match the available bandpass channel.

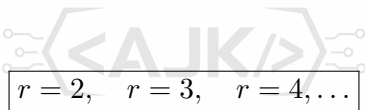
Multilevel ASK

Instead of only two amplitude levels, we can use:

4, 8, 16, ...

different amplitudes.

Therefore:


$$r = 2, \quad r = 3, \quad r = 4, \dots$$

bits can be transmitted at a time.

Note

Multilevel ASK is not implemented as pure ASK; it is implemented using QAM.

Frequency Shift Keying – FSK

Basic Idea

In FSK, the **frequency** of the carrier is varied to represent digital data.

- Frequency is constant during one signal element.
- Frequency may change for the next element.
- Amplitude remains constant.
- Phase remains constant.

Data → Frequency Changes

Binary FSK – BFSK

BFSK uses two carrier frequencies:

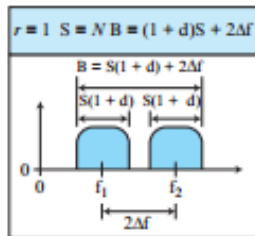
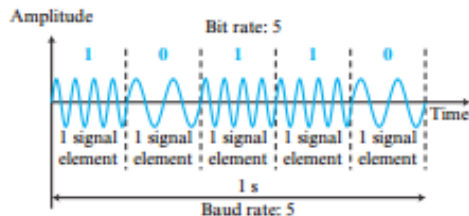
f_1	f_2
-------	-------

- f_1 represents data element 0.
- f_2 represents data element 1.

The carrier frequencies are normally very high and their difference is relatively small.

Figure 7.37 – Binary FSK

Figure 7.37 Binary frequency shift keying



FSK Bandwidth

If the two carrier frequencies are separated by $2\Delta f$:

$$B = (1 + d)S + 2\Delta f$$

where:

- S = signal rate
- d = modulation/filtering factor
- $2\Delta f$ = separation between carrier frequencies

For proper operation:

$$2\Delta f \geq S$$

Important

FSK can be viewed as two ASK signals, each using its own carrier frequency.

Multilevel FSK – MFSK

More than two frequencies can be used.

For example:

4 frequencies $\rightarrow$ 2 bits at a time

8 frequencies $\rightarrow$ 3 bits at a time

In general, L frequencies can represent:

$$\log_2 L \text{ bits}$$

The frequencies need to be separated by $2\Delta f$.

For proper operation:

$$2\Delta f \geq S$$

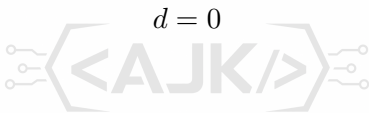
MFSK Bandwidth

For L different frequencies:

$$B = (1 + d)S + (L - 1)2\Delta f$$

If:

and:



A watermark logo is centered on the slide. It features the text 'ARSHA JK VJ CET PCCST501' in a stylized, bold font. The text is enclosed within a hexagonal frame that has circuit-like lines extending from its corners. Above the text, the equation $d = 0$ is written.

$$d = 0$$

$$2\Delta f = S$$

then:

$$B = S + (L - 1)S$$

$$B = LS$$

Phase Shift Keying – PSK

Basic Idea

In PSK, the **phase** of the carrier is varied to represent different signal elements.

- Amplitude remains constant.
- Frequency remains constant.
- Phase changes according to the data.

Data → Phase Changes

Important

PSK is more common than ASK or FSK.

Binary PSK – BPSK

BPSK uses two signal elements:

0°

180°

Advantage

BPSK is less susceptible to noise than ASK because noise can change amplitude more easily than phase.

Compared with BFSK

PSK does not require two separate carrier signals.

BPSK Bandwidth

The bandwidth of BPSK is:

$$B = (1 + d)S$$

It is the same as binary ASK but less than BFSK.

Why?

No bandwidth is wasted in separating two carrier frequencies.

Quadrature PSK – QPSK

QPSK sends:

2 bits per signal element

It uses two separate BPSK modulations:

In-phase

Quadrature

The two carriers are separated by:

90°

QPSK Operation

Incoming bits first undergo serial-to-parallel conversion.

Input Bits $\rightarrow$ Serial-to-Parallel

Then:

One bit $\rightarrow$ In-phase BPSK

Next bit $\rightarrow$ Quadrature BPSK

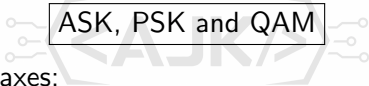
If the incoming bit duration is T , each BPSK signal has duration:

$$2T$$

Thus each BPSK signal has half the frequency of the original bit stream.

Constellation Diagram

A constellation diagram represents each signal element as a point. It is useful for:



ASK, PSK and QAM

The diagram has two axes:

- X-axis = in-phase component
- Y-axis = quadrature component

Reading a Constellation Diagram

For every point, four pieces of information can be obtained.

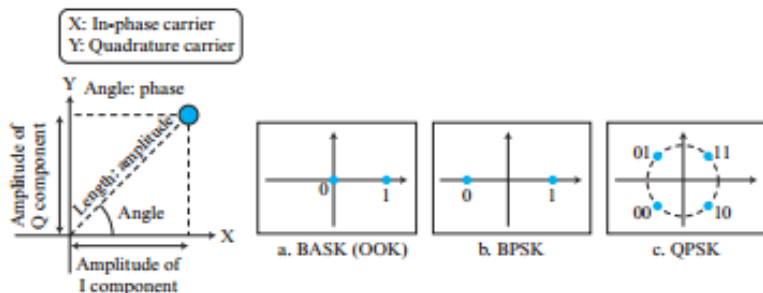
- ① Projection on X-axis: peak amplitude of in-phase component.
- ② Projection on Y-axis: peak amplitude of quadrature component.
- ③ Distance from origin: peak amplitude of the complete signal element.
- ④ Angle with X-axis: phase of the signal element.

Use

Constellation diagrams provide amplitude and phase information at a glance.

Figure 7.39 – Constellation Diagrams

Figure 7.39 Concept of a constellation diagram

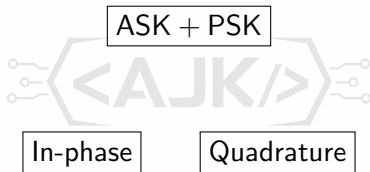


Quadrature Amplitude Modulation – QAM

PSK is limited by the ability of equipment to distinguish small phase differences.

QAM combines:

It uses two carriers:



with different amplitude levels for each carrier.

Important

QAM is the dominant method of digital-to-analog modulation.

QAM Variations

QAM can have many variations.

Examples include:

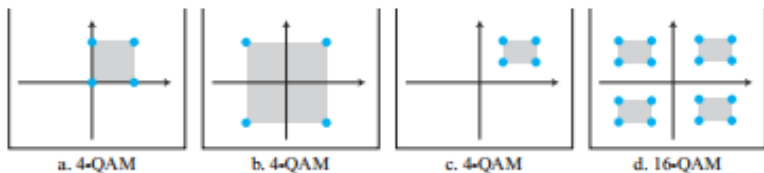
- 4-QAM using unipolar NRZ.
- 4-QAM using polar NRZ.
- 4-QAM with two positive amplitude levels.
- 16-QAM using multiple positive and negative levels.

Special Case

4-QAM using polar NRZ is exactly the same mechanism as QPSK.

Figure 7.40 – QAM Constellations

Figure 7.40 Constellation diagrams for some QAMs



QAM Bandwidth

The minimum bandwidth required for QAM is the same as that required for ASK and PSK.

$$B_{\text{QAM,min}} = B_{\text{ASK}} = B_{\text{PSK}}$$

Advantage

QAM combines the advantages of PSK and ASK.

Digital-to-Analog Conversion – Summary

Technique	Changed Characteristic
ASK	Amplitude
FSK	Frequency
PSK	Phase
QAM	Amplitude + Phase

Remember

QAM combines ASK and PSK and is the commonly used efficient digital-to-analog modulation method.

7.3.2 Analog-to-Analog Conversion

Definition

Analog-to-analog conversion represents analog information using an analog signal.

Why Modulate an Analog Signal?

The information is already analog. Modulation is required when the medium is bandpass or only a bandpass channel is available.

Example: Radio

Each radio station produces a low-pass signal, but different stations must occupy different frequency ranges.

Analog Modulation Techniques

Analog-to-analog conversion can be performed using:

AM FM PM

where:

- AM = Amplitude Modulation
- FM = Frequency Modulation
- PM = Phase Modulation

FM and PM are usually categorized together.

Amplitude Modulation – AM

In AM, the amplitude of the carrier follows the changing amplitude of the modulating signal.

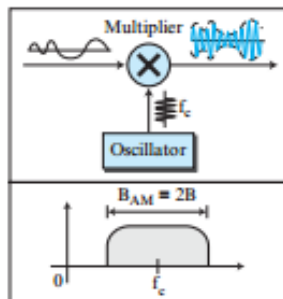
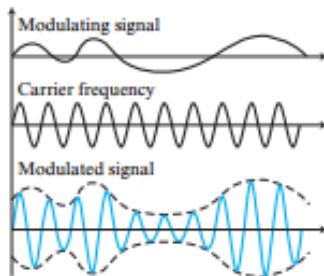
- Carrier frequency remains constant.
- Carrier phase remains constant.
- Carrier amplitude changes.

The modulating signal forms the **envelope** of the carrier.

Information → Carrier Amplitude

Figure 7.41 – Amplitude Modulation

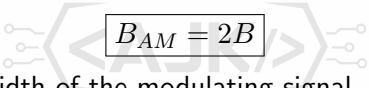
Figure 7.41 *Amplitude modulation*



AM Bandwidth

AM can be implemented using a simple multiplier because carrier amplitude must change according to the amplitude of the modulating signal.

The modulation produces a bandwidth:


$$B_{AM} = 2B$$

where B is the bandwidth of the modulating signal.

The bandwidth is centered on the carrier frequency.

Important

The upper and lower components carry the same information. Therefore, some systems discard one half and reduce the bandwidth by half.

Frequency Modulation – FM

In FM, the frequency of the carrier follows the changing amplitude of the modulating signal.

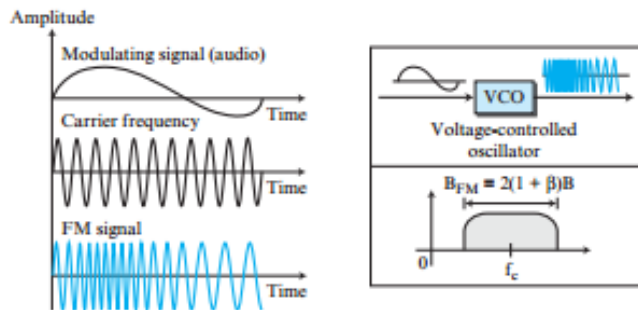
- Carrier amplitude remains constant.
- Carrier phase remains constant.
- Carrier frequency changes.

Information → Carrier Frequency

FM is normally implemented using a **voltage-controlled oscillator (VCO)**.

Figure 7.42 – Frequency Modulation

Figure 7.42 Frequency modulation



FM Bandwidth

The actual bandwidth of FM is difficult to determine exactly.
An empirical expression is:

$$B_{FM} = 2(1 + \beta)B$$

where:

- B = bandwidth of the modulating signal
- β = factor depending on modulation technique

A common value is:

$$\beta = 4$$

Phase Modulation – PM

In PM, the phase of the carrier follows the changing amplitude of the modulating signal.

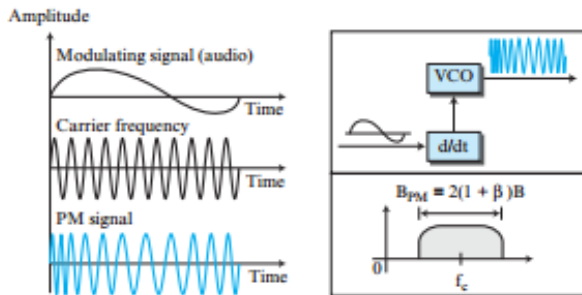
- Carrier amplitude remains constant.
- Carrier frequency remains constant.
- Carrier phase changes.

Information $\rightarrow$ Carrier Phase

PM can be implemented using a voltage-controlled oscillator along with a derivative.

Figure 7.43 – Phase Modulation

Figure 7.43 *Phase modulation*



FM versus PM

FM and PM are mathematically related.

FM

The instantaneous change in carrier frequency is proportional to the amplitude of the modulating signal.

PM

The instantaneous change in carrier frequency is proportional to the derivative of the amplitude of the modulating signal.

$FM \rightarrow \text{Amplitude}$

$PM \rightarrow \text{Derivative of Amplitude}$

PM Bandwidth

The actual PM bandwidth is difficult to determine exactly. It can be several times the bandwidth of the analog signal. An empirical expression is:

$$B_{PM} = 2(1 + \beta)B$$

The value of β is lower for PM:

$$\beta \approx 1$$

for narrowband PM, and

$$\beta \approx 3$$

for wideband PM.

Analog Transmission – Complete Summary

Digital-to-Analog

- ASK – amplitude changes
- FSK – frequency changes
- PSK – phase changes
- QAM – amplitude + phase

Analog-to-Analog

- AM – amplitude changes
- FM – frequency changes
- PM – phase changes

Important Formulas

$$B_{ASK} = (1 + d)S$$

$$B_{FSK} = (1 + d)S + 2\Delta f$$

$$2\Delta f \geq S$$

$$B_{MFSK} = (1 + d)S + (L - 1)2\Delta f$$

For $d = 0$ and $2\Delta f = S$:

$$B_{MFSK} = LS$$

$$B_{AM} = 2B$$

$$B_{FM} = 2(1 + \beta)B$$

Quick Revision

- 1 Why is analog transmission required for a bandpass channel?
- 2 What is digital-to-analog conversion?
- 3 How does ASK represent data?
- 4 What is the difference between BFSK and MFSK?
- 5 Why is PSK less susceptible to noise than ASK?
- 6 How does QPSK transmit two bits at a time?
- 7 What is a constellation diagram?
- 8 Why is QAM efficient?
- 9 Why is analog modulation needed for radio?
- 10 Compare AM, FM and PM.

Key Takeaways

Remember

ASK $\rightarrow$ Amplitude

FSK $\rightarrow$ Frequency

PSK $\rightarrow$ Phase

QAM $\rightarrow$ Amplitude + Phase

AM $\rightarrow$ Amplitude

FM $\rightarrow$ Frequency